

TRACE-RAF: Trust-Gated Residual Analog Correction with Evidence-Preserving Retrieval for Auditable PM2.5 Forecasting

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Abstract—Pollutant persistence, meteorology, calendar effect, scarce environmental events and temporal distribution shift are all factors that need to be considered in short-horizon PM2.5 forecasting. When there is already a good local model (predictable part), it can be damaging to have a full analogue future retrieved by existing retrieval-augmented forecasters which can provide similar trajectories. In this study, TRACE-RAF, a Trust-gated Residual Analog Correction mechanism integrated in an auditable forecasting framework that evaluates event context as auxiliary evidence is proposed. A Source-Audited Context Builder brings alignment of EPA’s official PM2.5 data with the NOAA GHCN weather, calendar, and hash-verified NOAA Storm Events data. The Leakage-Safe Event-Context Retriever imposes a strict embargo on candidates, while TRACE-RAF retrieves the residual trajectories, instead of raw targets, that are out-of-fold. Each residual is scaled by an 18-variable reliability gate, which, based on retrieval quality, component disagreement, correction spread, event context, and drift evidence, increases or decreases the weight of the residual. Evaluation is done with a 168 hour lookback window, 24 hour horizon, 45,784 origins in the training set, 7,950 origins in the validation set, and a fixed final holdout of 2,460 origins in Los Angeles County between 2019 and 2025. TRACE-RAF obtains MSE 40.062, MAE 4.134, and RMSE 6.329. It is 4.63% better than the XGBoost context model and within 0.07% of the stronger LightGBM context model, with MSE 39.144 for the Ridge regression (best overall). The TRACE-RAF decreases the MAE by 0.021 in comparison to the base ensemble, which is a nominal effect that is not significant after Holm correction. Event-conditioned and event-free TRACE-RAF variants are almost the same, while event-conditioned output-space fusion significantly worsens the frozen Chronos-Bolt-small MAE after multiplicity adjustment. Results show selective residual correction and an auditable evaluation protocol are supported, but they do not establish an event-conditioned gain or a general conclusion about retrieval-augmented TSFMs. Storm Events will continue to produce retrospective sensitivity analyses since it does not have machine-readable publication timings.

Index Terms—Distribution-shift diagnostics, auditable forecasting, time-series foundation models, PM2.5, retrieval-augmented forecasting, residual correction, air quality forecasting.

I. INTRODUCTION

Because fine particulate matter (PM2.5) has respiratory and cardiovascular health impacts [1] it is a significant urban air-quality hazard. Good quality short-range PM2.5 forecasts can be used to aid public-health warnings, exposure minimisation, transport planning and environmental management. However, in Los Angeles County PM2.5 is not easily modeled based on

pollutant history. Changes in concentration may occur due to wind speed, boundary-layer conditions, precipitation, seasonal variations, holidays, wildfire smoke and other unusual storm or high wind events. All these factors render the forecasting problem “temporal,” and “contextual.”

Pre-training TSFMs on vast time-series corpora and transferring the forecasting capability to new domains has recently been proposed as a way to enhance the generalization of forecasting. Examples are the TimesFM, Moirai, MOMENT, Chronos and Tiny Time Mixers [2]–[6]. At the same time, retrieval-augmented generation (RAG) has demonstrated that non-parametric memory can be useful to foundation models for retrieving relevant external knowledge at inference time [7], [8]. TimeRAF brings in these two concepts to zero-shot time-series forecasting by retrieval from a pre-selected curated knowledge base and insertion in a frozen TSFM via Channel Prompting [9].

One domain specific retrieval question, not completely solved by numerical time-series retrieval is air-quality forecasting. A historical PM2.5 window can be similar to the current window, but under different weather/effect conditions. On the other hand, a window that has moderate numerical similarity may provide more information if it has high wind or precipitation or holiday or stagnation or smoke context. This encourages a retrieval system that can incorporate source information records, weather, calendar data and the distribution-shift indicator as part of the evidence, not as the metadata.

This study proposes TRACE-RAF, a Residual Analog Correction mechanism for auditable PM2.5 forecasting with event context evaluated as an auxiliary evidence channel under temporal distribution shift. The framework involves retrieval augmentation in an environmental forecasting scenario without being architecturally similar to that of TimeRAF’s learned retriever and Channel Prompting mechanism. The four implemented modules are a Source-Audited Context Builder (SACB), a Leakage-Safe Event-Context Retriever (LSER), the TRACE residual-correction core and a Drift-Aware Forecast and Evidence Head (DFEH). They work together to answer four questions: how to put official records into retrieval-ready context; how to select candidates without “temporal leakage”; how analogue residuals can correct a good supervised forecast without being a substitute for it; how each forecast can retain evidence of retrieval, events, drift, and uncertainty.

These are the main contributions:

- Source-Audited Context Builder: a replicable procedure that is used to link EPA PM2.5, NOAA weather, calen-

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dar and hash-validated NOAA Storm Events records to maintain source identity, source coverage checks, and an explicit assumption of event-availability.

- Leakage-Safe Event-Context Retriever: a historical analogue mechanism integrating similarity in both PM2.5 shape and weather, calendar and event similarity after applying an embargo that every candidate target must end prior to the query lookback starting.
- An out-of-fold residual memory and validation-trained reliability gate to selectively apply analogue corrections to a convex LightGBM–XGBoost context ensemble: TRACE residual correction.
- Drift-Aware Forecast and Evidence Head: a layer of evaluation that provides direct, residual-corrected and frozen paths of Chronos-Bolt, transparent shift indicators, probabilistic diagnostics and machine-readable forecast evidence under one chronological protocol.

Evaluation is performed with a final chronological hold-out and comprises linear, boosted tree, neural, retrieval-only, placebo-fusion and frozen-TSFM baselines. Paired inference is based on 168-hour blocks, 5,000 bootstrap resamples, HAC-corrected Diebold–Mariano tests and Holm adjustment over the declared comparison family. This design will ensure that negative evidence is as apparent as positive evidence and that event association is not inferred as causation.

II. LITERATURE REVIEW AND RELATED WORK

A. Statistical, Machine-Learning, and Deep Forecasters

There are various forecasting methods that vary in their trade of inductive bias, data required and computational effort. Autoregressive and/or seasonal models continue to be useful models as baselines since their failure modes are readily examined. This is extended to nonlinear lag, rolling, weather, and calendar interactions with the tree ensembles like XGBoost and LightGBM without the need for a sequence encoder [10], [11]. RNNs with LSTM or GRU cells, however, are able to learn stateful temporal representations [12], [13]. However, systematic studies warn that recurrent networks are not straightforward alternatives to good statistical baselines, and must be well-preprocessed and evaluated [14]. DeepAR generalizes the recurrent model to also forecast a probability over related series [15]. Further large benchmark archives such as the Monash Forecasting Repository enable the reproducible comparison of forecasts in different classes of time-series domains [16]. Transformer families focus on dependencies of longer timeframes: Informer reduces the attention cost, Autoformer and FEDformer decompose the temporal variations, Crossformer and iTransformer model cross-variable structure, while TimesNet reorganizes temporal variation into two dimensional patterns [17]–[22]. The achievement of these gains does not necessarily mean that the level of architectural complexity needs to be increased. Simple decomposition or patch-based inductive biases are seen to be very competitive in DLinear and PatchTST [23], [24] while a multiscale decomposition and mixing is used in TimeMixer [25]. Seasonal and boosted tree baselines are thus retained in the evaluation, not taking a deep backbone as the default winner. This option

is in line with recent survey articles in the literature that state that model selection should be based on horizon and covariate structure, uncertainty requirements, and data size and scale, and not solely on architecture novelty [26], [27].

B. Time-Series Foundation Models

To avoid training a separate representation of each series, time-series foundation models (TSFMs) aim to transfer knowledge across datasets. Decoder-only forecasting design (TimesFM), universal forecasting Transformer (Moirai), multiple time-series forecasting tasks (MOMENT), value as token (Chronos), and small zero/few-shot transfer (Tiny Time Mixers) [2]–[6]. Timer, Timer-XL, UniTime and Time-LLM investigate the concepts of generative pretraining, long-context scaling, cross-domain unification, or language-model reprogramming [28]–[31]. Frozen pretrained models have also been tried as general computation engines in the time series analysis field [32]. All these studies inspire zero-shot evaluation, and provide evidence that transfer performance is related to the representation of the input and the shift of domains. Thus, in the present study, we compare the performance of frozen Chronos-Bolt to that of locally supervised models on a common set of forecast origins, instead of taking the superiority of the foundation models for granted.

C. Retrieval-Augmented Forecasting

In retrieval augmentation, the parametric knowledge is detached from an external memory. RAG and Dense Passage Retrieval are the early works that set the example of retrieve-then-condition in the field of natural-language processing [7], [8]. More recently, the issue of retrieval has been explored in the context of diffusion forecasting and assisted generation workflows in time series [33], [34]. TimeRAF is the most relevant methodologically: it builds a time-series knowledge base, trains an MLP dual-encoder retriever, picks out the top- k candidates and incorporates the embedding of candidates in a frozen TSFM via Channel Prompting [9]. Its ablations split retrieval quality, integration, number of candidates, the composition of the knowledge base and backbone selection.

The current framework is also based on the principle of external memory, but it tackles another experimental question. It is based on transparent random, cosine and event-context similarity rather than a learned dual encoder. The main path of TRACE-RAF is to retrieve the historical model residuals and use a learnt trust gate; the other arm of TRACE-RAF, called Chronos-Bolt, is used for output space fusion but not for internal Channel Prompting. This therefore is not a copy of TimeRAF. An audited environmental adaptation to test the effectiveness of contextual analogues in providing additional information following a strong local forecast.

D. Air-Quality Forecasting and Official Context

The pollutants history with temperature, humidity, wind, precipitation and pressure are commonly used as variables to prepare PM2.5 forecasting. Recent reviews show that there has been a paradigm change from single-station recurrent

models to attention-based, hybrid and spatiotemporal models [35], [36]. For instance, AirFormer performs spatiotemporal modeling in a deterministic way and uncertainty estimation in a stochastic manner at the nation level [37]. Multi-station designs capture the spatial dependence, while the current study considers an aggregate at the county level, and the source of context evidence. EPA AirData provides PM2.5 observations that are regulated, NOAA GHCN provides observations for hourly meteorology, and NOAA Storm Events provides structured hazard records [38]–[40]. Smoke evidence could be introduced in NOAA Hazard Mapping System products but they were not reported in the run [41].

E. Distribution Shift and Forecast Evidence

Temporal nonstationarity can affect mean, variance and relationship between inputs and future values. Adaptive normalization methods react to this issue by calculating statistics on local temporal scales [42]; TRACE-RAF makes use of interpretable shift indicators to audit regimes but without claiming to have removed shift. There are also variations in the scope of explanation methods. SHAP can assign predictions to model inputs [43] and information-theoretic saliency can aid the counterfactual analysis of probabilistic forecasts [44]. The narrower evidence layer implemented includes signed XGBoost effects, retrieved cases, event identifiers, drift components and an uncertainty proxy. The records provide information on traceability, and do not prove causes of the weather or events.

III. CRITICAL GAPS AND LIMITATIONS OF PREVIOUS STUDIES

Four gaps in the event-sensitive urban air-quality forecasting are identified from previous studies. First, temporal forecasters and numerical retrieval systems do not always maintain the assumptions about provenance, coverage and availability that are made by local contextual records. A numerically similar PM2.5 window could be associated with a different weather or event, and the information associated with a retrospective event could be incorrect for forecast time. The SACB fills this void by bringing official records into the equation under explicit source, coverage and availability audits.

Second, if a candidate answer (either an input or a target) is present in the context of the query, historical retrieval may add "look-ahead bias. Just because it is similar, doesn't mean it can't be the same future in the retrieved evidence! The LSER tackles this deficit by imposing a strict temporal embargo and the training-history restriction, and then four separate terms of PM2.5 similarity, weather similarity, calendar similarity, and event similarity that can be audited independently.

Third, information provided by raw analogue futures can replicate or contradict that of a good supervised model. TRACE-RAF fills this void by saving only out-of-fold residual trajectories and training an origin level trust gate with an origin-level validation-only correction utility. Thus it does not pose the question of whether an analogue raw future should supplant the base forecast but asks whether an analogue explains the rest of the error.

Fourth, retrieval gains are reported without determining if they withstand distribution shift, simple forecast-averaging placebos, or can be attributed to stored evidence. To fill this void, the DFEH analyzes direct, residual-corrected, and frozen Chronos-Bolt paths on the same common origins, displays indicators of the shifts, and preserves evidence of the features, retrieval, events and uncertainties that are machine-readable. This component helps in comparison and auditability but not in providing causal explanation.

These maps are the boundaries of novelty of this study. TRACE-RAF does not include a learned dual-encoder retriever or internal Channel Prompting and the present experiment in a single county does not afford geographic generalization.

IV. CORE CONTRIBUTIONS

The proposed framework offers four mechanisms which are in one-to-one relationship with the gaps above:

- **Source-Audited Context Builder (SACB)**: builds an aligned version of the context with 85 dimensions, based on official records of PM2.5, meteorological, calendar, and events, and preserving metadata of sources and availability.
- **Leakage-Safe Event-Context Retriever (LSER)**: only allows retrieval of training-history analogues after enforcing a candidate-target embargo, and exposes the contribution of each similarity channel, and retrieved trajectory evidence.
- **Trust-gated Residual Analog Correction (TRACE)**: builds an embargoed memory of forecast out-of-fold residuals, scales the retrieved corrections to the query scale and uses a reliability gate based on validation, instead of replacing all forecasts with residuals.
- **Drift-Aware Forecast and Evidence Head (DFEH)**: combines all the above within one chronological evaluation contract.

Qualifications are made for the empirical contribution on purpose. TRACE-RAF improves on its context ensemble in nominal MAE inference and remains competitive with the best nonlinear baseline, but the family-wise Holm corrected component comparison revealed no verified all-origin effect for the event channel.

V. PROBLEM FORMULATION

We will use x_t to represent the PM2.5 reading at time t . The historical air-quality lookback window for forecast time t is defined as in equation (1).

$$X_t = [x_{t-L+1}, x_{t-L+2}, \dots, x_t], \quad (1)$$

The resulting lookback length, L , is where W_t as weather covariate corresponding to the lookback window, C_t as calendar feature and E_t as context of events at the origin of the forecast in a recorded source assumption. The target of Equation (1) is defined as in Equation (2).

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \dots, x_{t+H}], \quad (2)$$

where H is the forecasting period. Equation (2) thus represents the 24-step vector, which is calculated at each valid origin.

TABLE I
MAIN SYMBOLS USED IN THE PROPOSED FORMULATION

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
E_t	Event context available at the forecast origin
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event knowledge
D_t	Drift indicators
q_t^B	Non-event base context in $\mathbb{R}^{46}$
q_t	SACB context vector in $\mathbb{R}^{85}$
G_t	LSER top- k retrieved set
$\rho(G_t)$	Retrieval summary in $\mathbb{R}^{51}$
$\hat{Y}_t^B$	Convex context-ensemble forecast
Δ_t	Retrieved residual correction in $\mathbb{R}^H$
z_t	TRACE reliability vector in $\mathbb{R}^{18}$
g_t	Validation-calibrated TRACE gate in $[0, 1]$
$\mathcal{O}_t$	Machine-readable DFEH evidence record
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

The forecasting function is given in (3).

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \quad (3)$$

Here, R_t^{ts} is the time-series knowledge being retrieved, R_t^{ev} is the event knowledge being retrieved and D_t are the distribution-shift indicators. The target setting that has been implemented is as follows: $L = 168$ hours and $H = 24$ hours. The complete information interface is given by equation (3); which of these modules are implemented to create these arguments is given in the next section.

The time based split is given in equation (4).

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \quad (4)$$

where training origins span January 8, 2019–August 23, 2024, validation origins span August 25, 2024–August 23, 2025, and test origins span August 25, 2025–December 30, 2025. These defined thresholds result in 45,784, 7,950 and 2,460 windows, respectively. Ordering in (4) is used also for the feature fitting, retrieval eligibility and model selection and final evaluation; the test partition is not used to maintain a nominal percentage.

The notation essential to the understanding of the tables is summarized in Table I.

VI. METHODOLOGY

TRACE-RAF comes as 4 inter-related modules. The Source-Audited Context Builder (SACB) takes heterogeneous official records, organizes them chronologically by the hour and preserves provenance and availability metadata. The Leakage-Safe Event-Context Retriever (LSER) builds up an embargoed historical memory and retrieves the analogues based on PM2.5, weather, calendar, and event similarity. The TRACE core transforms out-of-fold errors made by a supervised context ensemble into an analogue residual memory and establishes a notion of when to rely on that residual memory. The Drift-Aware Forecast and Evidence Head (DFEH) reveals the final forecast, the comparison paths, shift indicators and a machine-readable evidence record. No image operator, no internal

TSFM Channel Prompting or Test Time Parameter Fitting (TTPF) is assumed for all the modules, which are based on one-dimensional time series or tabular features.

A. Framework Overview and Forward Pass

The entire dataflow is indicated in figure 1. The first phase is to load, audit and transform the EPA PM2.5 data and NOAA GHCN, NOAA weather, calendar and NOAA Storm Events data into the SACB to form query contexts and input–target windows. The LSER takes the historical embargo first and then ranks those who meet eligibility requirements. Meanwhile, expanding window XGBoost and LightGBM models are built and strictly out-of-fold residuals are produced for the training memory. TRACE-RAF is able to retrieve and align those residuals, and estimate their reliability, while adding a gated correction to the context ensemble. The forecast, retrieval cases, event identifiers, drift evidence and uncertainty fields are recorded by the DFEH, while frozen Chronos-Bolt is an evaluated separately benchmarked path.

The entire mapping for every forecast origin t is given by equation (5).

$$\hat{Y}_{t+1:t+H} = F_{\theta,\psi}(\phi(X_t, W_t, C_t, E_t), G_t, D_t, \mathcal{R}), \quad (5)$$

Here, $\phi(\cdot)$ is the tabular feature construction, G_t the retrieved historical evidence, D_t the drift evidence and $\mathcal{R}$ the out-of-fold residual memory. The one methodological limitation is that all the candidate targets and all the residuals in G_t and $\mathcal{R}$ terminate before the beginning of the lookback window for the query. The forward pass below is the expanded form of (5).

The implemented forward pass is extended to tensor notation in equation (6), which is for a batch of B origins.

$$\begin{aligned}
Q^B &= \text{SACB}_B(X, W, C) \in \mathbb{R}^{B \times 46}, \\
Q &= [Q^B \parallel E] \in \mathbb{R}^{B \times 85}, \\
G^r &= \text{LSER}(Q, \mathcal{R}) \in \mathbb{R}^{B \times 8 \times 24}, \\
(\hat{Y}^L, \hat{Y}^X) &= \mathcal{H}(Q^B, C^+) \in (\mathbb{R}^{B \times 24})^2, \\
\hat{Y}^B &= \eta \hat{Y}^L + (1 - \eta) \hat{Y}^X \in \mathbb{R}^{B \times 24}, \\
(\Delta, Z) &= \mathcal{A}(G^r, \hat{Y}^B, D) \in \mathbb{R}^{B \times 24} \times \mathbb{R}^{B \times 18}, \\
g &= \lambda \text{clip}(h_\psi(Z), 0, 1) \in [0, 1]^{B \times 1}, \\
U &= \hat{Y}^B + g \odot \Delta \in \mathbb{R}^{B \times 24}, \\
(\hat{Y}, \mathcal{O}) &= \text{DFEH}(U, D, G^r) \in \mathbb{R}^{B \times 24} \times \mathcal{E}.
\end{aligned} \quad (6)$$

Here, $\eta = 0.6521$ is the out-of-fold-selected LightGBM weight, Δ is the aligned residual correction, Z are 18 features that are target free and $\lambda = 0.25$ is the validation-selected correction strength. $\mathcal{E}$ is the schema of evidence record $\mathcal{O}$. $\mathcal{H}$ and $\mathcal{A}$ are the paired context heads and residual-alignment stage. The 23 PM2.5 and 14 weather features and nine calendar features are in the Q^B part, while the 39 event features are in the Q part, and are not present in either base head. 9 known future-calendar terms are added to Q^B for each horizon, which provides a total of 55 inputs to each base regressor. The feature-level M09 and frozen Chronos-Bolt arms remain a controlled comparison, and are not hidden within TRACE-RAF as per equation (6).

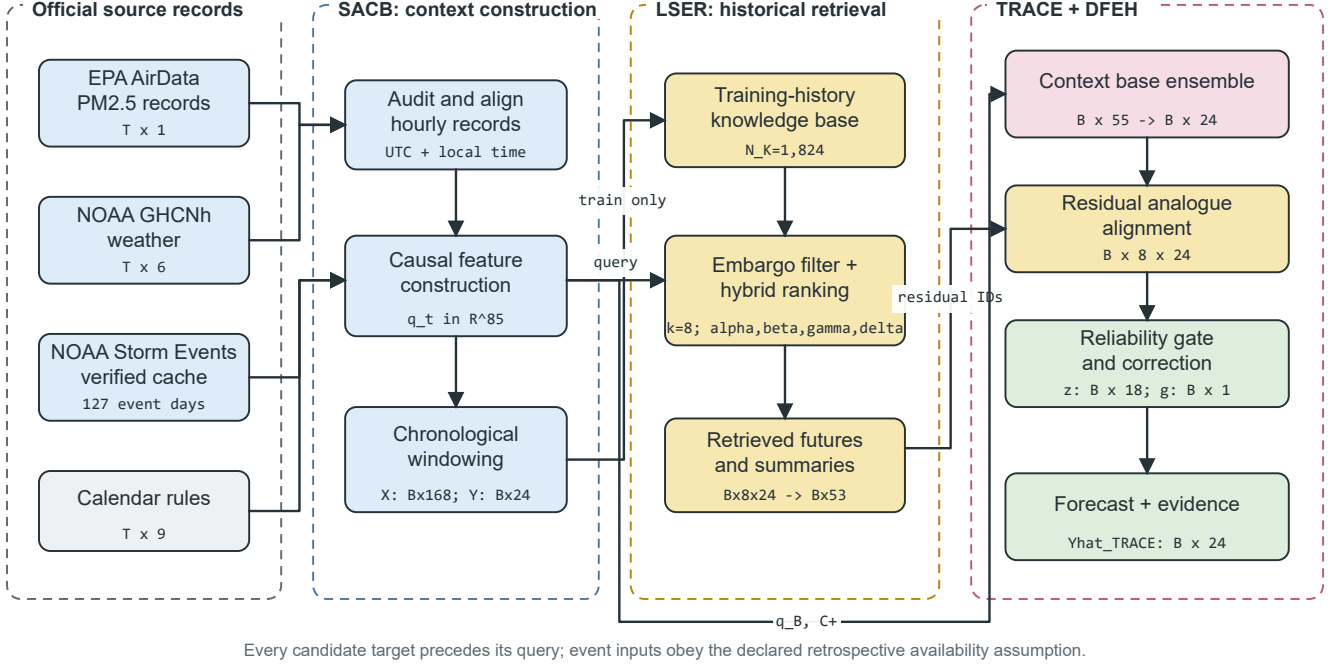


Fig. 1. Proposed TRACE-RAF pipeline. Official records pass through SACB, training-history candidates are filtered and ranked by LSER, out-of-fold residual analogues are scaled by the TRACE trust gate, and DFEH produces the 24-hour forecast with traceable evidence. Tensor annotations show the manifest-backed configuration.

B. Source-Audited Context Builder

The motivation behind the SACB is to maintain the data provenance and temporal availability before model fitting. As can be seen in Figure 2, there are checks before the construction of the feature, which are specific to the source. EPA monitor records are consolidated into an hourly county median, NOAA weather data is synced from the weather surface station of choice, and event archives are hash checked to a manifest and calendar data is calculated from the local time stamp. The context is then windowed and broken down into chronological parts; no interpolation of target values.

1) *Dataset Construction:* Los Angeles County (LAC) hourly PM2.5 data is collected from the US EPA AirData/AQS parameter 88101 [38]. If only one site meets the coverage criterion, then the data audit uses the same site for coverage and continuity. Since none of the monitors met this criteria the target is the hourly median of the monitors on the official Los Angeles County AQS network. Timestamps are converted to America/Los_Angeles, daylight saving explicitly taken care of. Gaps in the input are only filled with observations from the past, in the case of a short gap no target is interpolated. The weather covariates are obtained from the closest appropriate NOAA Global Historical Climatology Network hourly (GHCNh) weather station [39]. Calendar terms include cyclic hour, weekday, and month names, weekend, US Federal holidays, and CA holidays.

The concatenation of the feature groups at each origin is

given by equation (7).

$$\begin{aligned} q_t^B &= [p_t \parallel w_t \parallel c_t] \in \mathbb{R}^{23+14+9} = \mathbb{R}^{46}, \\ q_t &= [p_t \parallel w_t \parallel c_t \parallel e_t] \in \mathbb{R}^{23+14+9+39} = \mathbb{R}^{85}. \end{aligned} \quad (7)$$

where p_t , w_t , c_t and e_t are the PM2.5, weather, calendar and event feature vectors, respectively. Fixed lags, rolling moments and extrema, differences, current value and fill flag are included in the PM2.5 group. The weather group consists of 6 measured variables, a missing-flag for precipitation, flagged variables for the weather condition, a stagnation-flag and 24-hour rolling means of the weather variables. The event group includes one hour and active counts, 24/72/168-hour counts, category counts (72 hours) and event-burst ratio. Therefore, Equation (7) sets both feature orderings; that of 46 dimensions (base context) for C01, M04, C05 and C06 and that of 85 dimensions (event-aware context) for LSER and event-conditioned controls.

Normalisation statistics are only fit on the training split. To prevent mixing of different scales affecting similarity, query and candidate windows are preprocessed in the same manner. The reported run contains 56,194 valid windows with $X \in \mathbb{R}^{56194 \times 168}$ and $Y \in \mathbb{R}^{56194 \times 24}$. Out of these, 45,784 are in the training set, 7,950 in the validation set and 2,460 are in the final test holdout set.

C. Leakage-Safe Event-Context Retriever

The LSER distinguishes between eligibility over time and relevance. The embargo is first displayed in figure 3, a candidate is considered only if their complete target is before the

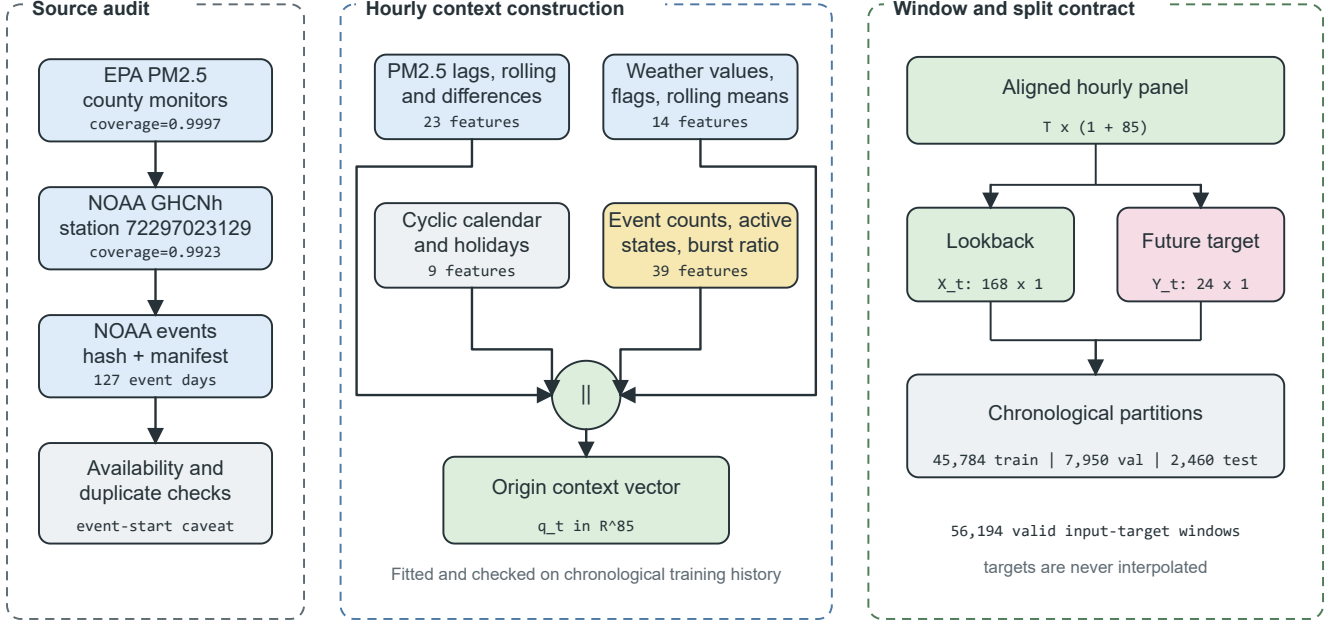


Fig. 2. Source-Audited Context Builder. Source checks feed four feature groups that form an 85-dimensional origin vector. The aligned panel is converted into 168-hour inputs and 24-hour targets before the chronological split.

start of query lookback. Then the list of eligible candidates is compared on four similarity channels which are separately stored. The futures that are selected correspond to the scale of the query, and are summarized for the forecasting head. This ordering avoids that a high similarity score is sufficient for this temporal constraint to be ignored.

1) *Time-Series Knowledge Base*: Historical windows of PM2.5 are available in the time-series knowledge base. All the information in each record includes a window identifier, start time, end time, input and future values, weather summary, calendar summary, normalized window vector, split, and normalization identifier. The Primary Knowledge Base is 24-hour origin stride and has 1,824 training windows. There are 6-hour and 1-hour strides in sensitivity arms, which give 7,608 and 45,784 candidates, respectively. The query-specific embargo in Equation (9) is not affected by candidate overlap, only the memory density.

Formally the knowledge base is defined as in equation (8).

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^N, \quad (8)$$

where u_i is a candidate origin, X_i and Y_i are candidate lookback and future windows, m_i describes a weather summary, c_i a calendar summary, and e_i an event-context summary provided at u_i . The temporally eligible set of a query origin t is defined in equation (9).

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1, i \in \mathcal{T}_{train}\}, \quad (9)$$

To validate and test queries. The strict inequality in (9) makes sure that the target period of a candidate ends before the query input period starts. Training-history condition (9) is also taken

into account for the final validation and the test retrieval. The stored candidate schema is defined in Equation (8) and the admissible subset at query time is defined in Equation (9).

(10) is the time-series retriever's result for query window X_t .

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \dots, r_{t,k}^{ts}\}, \quad (10)$$

where k is number of retrieved candidates). The default is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$. Equation (10) is the returned records prior to summarizing future records.

2) *Event Knowledge Base*: Event knowledge base includes source-preserving information pertaining to Los Angeles County. The required structured source is NOAA Storm Events and the optional extension (not measured) is NOAA Hazard Mapping System fire and smoke records [40], [41]. Event identifier, start, end, information-availability assumption, location, category, source, source URL, title and summary are stored fields. Reported records include data for the following weather categories: wildfire, high wind, heavy rain, flood, and others; the data for the missing weather categories in the records are all zeros.

Model inputs include events counted every hour, events counted over a 24-, 72- and 168-hour window, events counted by category over a 72-hour window, events active and an event-burst ratio. The forecast evidence consists of the recent source identifiers. Target-horizon overlap is a post hoc subset label, and is never used as a model input. The publication time is not found in NOAA Storm Events and the implementation makes an assumption that the event start was available in a retrospective way. For strict operations experiments, records of real publication/issue time are required.

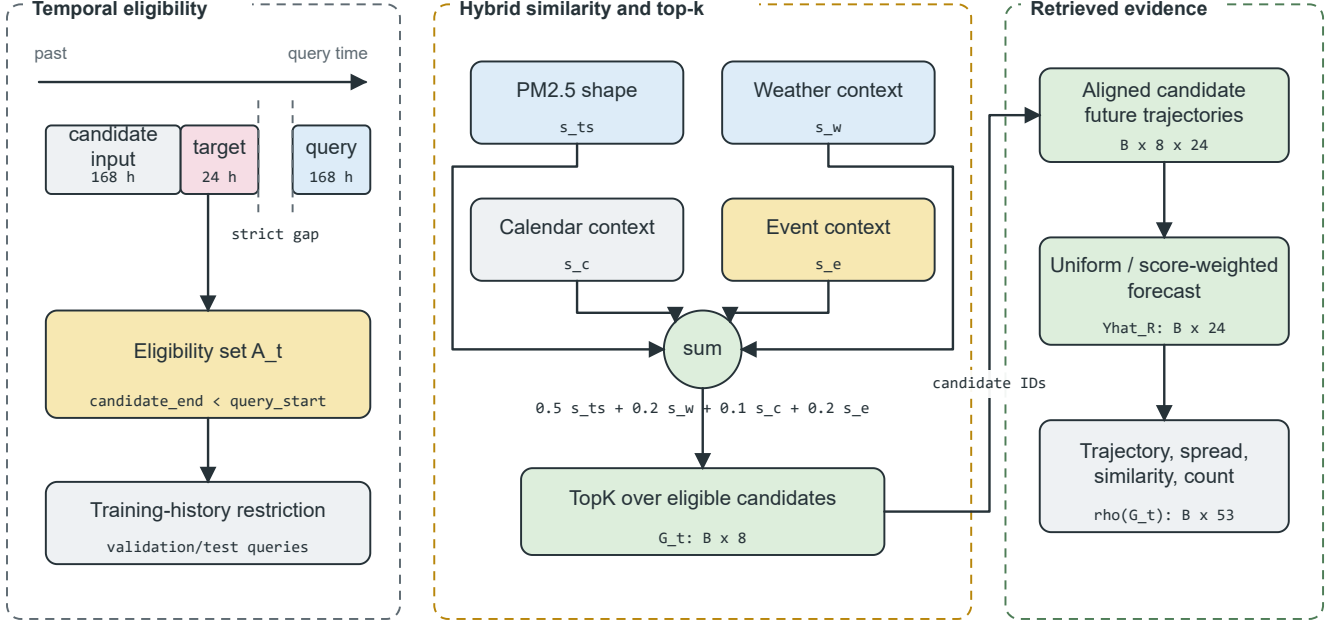


Fig. 3. Leakage-Safe Event-Context Retriever. A strict candidate-target embargo and training-history restriction are applied before hybrid ranking. The top eight candidates provide aligned future trajectories and a 53-dimensional retrieval summary.

3) *Hybrid Ranking and Future Alignment*: The query window and candidate window of PM2.5 data are mean-centered and scaled by their standard deviations and normalized to a unit length. Weather and calendar groups are standardized and mapped to $[0, 1]$ using the cosine similarity with candidate statistics. Explicit 0 vectors provided and event similarity clipped to 0-1. The hybrid retrieval score is represented by equation (11), which is a combination of time-series similarity, weather relevance, calendar relevance and event relevance.

$$s(q, i) = \alpha s_{ts}(q, i) + \beta s_w(q, i) + \gamma s_c(q, i) + \delta s_e(q, i), \quad (11)$$

The time-series similarity (s_{ts}), the weather similarity (s_w), the calendar similarity (s_c) and the event relevance (s_e) are all included. The implemented weights are $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$. To allow inspection of contribution from each relevance channel without a learned latent scoring function, Equation (11) maintains the explicitness of each relevance channel.

The following equation (12) is used to define a retrieved set.

$$G_t = \text{TopK}_{i \in \mathcal{A}_t} s(t, i). \quad (12)$$

$\mathcal{A}_t$ is applied to equation (12) and then the equation is evaluated. How candidate futures are restored to the query scale is defined in equation (13).

$$\tilde{Y}_{t,i} = \mu_t + \sigma_t \left(\frac{Y_i - \mu_i}{\max(\sigma_i, \epsilon)} \right), \quad i \in G_t, \quad (13)$$

where μ_t , μ_i are the lookback statistics for the query and candidate respectively and σ_t , σ_i are the respectively associated standard deviations. Retrieval-only forecasts consist of

the averaged $\tilde{Y}_{t,i}$ uniformly or by normalized scores that are all non-negative. The feature-level path preserves the mean trajectory, horizon-wise spread, mean and maximum time series similarity, and number of candidates, and is consequently a vector in $\mathbb{R}^{51}$, $\rho(G_t) \in \mathbb{R}^{51}$. The affine transformation that is expressed in Equation (13) will maintain the shape of the candidate trajectory while bringing the query back to its local scale.

D. Trust-Gated Residual Analog Correction

For TRACE-RAF, it's what you get back that changes. A raw candidate future is the future that is predictable based on the current context, plus the future that is not predictable based on the current context (the part the local model did not predict). Thus, if the future is added to a solid supervised forecast, the predictable structure can be double counted. TRACE-RAF, on the other hand, builds a memory of out-of-fold residual trajectories and generates trust coefficient of the retrieved correction at the origin level. This residual path from the two context models is reflected in the residual memory, trust gate, final forecast and evidence record as shown in figure 4.

1) *Out-of-Fold Base Forecasts and Residual Memory*: The base forecast is a convex combination of the LightGBM and XGBoost context models as shown in equation (14).

$$\hat{Y}_t^B = \eta \hat{Y}_t^L + (1 - \eta) \hat{Y}_t^X, \quad 0 \leq \eta \leq 1, \quad (14)$$

where η is only fitted from expanding window out of fold (OOF) training predictions. There are two OOF prediction blocks: $[0.50, 0.75)$ and $[0.75, 1.00)$ of the training sequence.

Each of the fitting targets stops before the first of the prediction inputs for each block. The audit records 22,728 and 34,154 fit origins, 11,446 prediction origins per block, and a passed embargo, in both cases.

The stored residual for the out-of-fold candidate i is given in equation (15).

$$\varepsilon_i = Y_i - \hat{Y}_i^{B, \text{OOF}} \in \mathbb{R}^H. \quad (15)$$

The residuals are individually horizon-clipped at 0.01 and 0.99 quantiles of the training only data set. At 24-hour stride, 909 candidates are selected for validation while a total of 3,795 and 22,892 residual candidates are validated at 6-hour and 1-hour strides respectively. The last step is selected prior to test retrieval.

2) *Residual Retrieval and Scale Alignment*: LSER scores the candidates for the residual memory in the same way as is done in Equation (9), based on the temporal eligibility. To aggregate the candidate residuals with the query scale, each $a_{t,i}$ is a non-negative retrieval weight, which is normalized, as shown in Equation (16).

$$\Delta_{t,h} = \sum_{i \in G_t} a_{t,i} \frac{\sigma_t}{\max(\sigma_i, \epsilon)} \varepsilon_{i,h}, \quad \sum_{i \in G_t} a_{t,i} = 1. \quad (16)$$

A spread vector is defined in a similar manner as the aligned residuals but is done on a horizon-wise basis. Unlike Equation (13), Equation (16) passes a history of the model error not a history of the target level.

3) *Reliability Gate*: An 18-dimensional target-free vector z_t is given for each origin, summarizing the mean, standard deviation and maximum of the base-forecast, the correction, the correction-spread, the mean and maximum of the retrieval similarity, the number of selected and eligible candidates retrieved in the log space, the selected-event fraction, whether an event conditioning was used, the mean and maximum disagreement between LightGBM and XGBoost, and the continuous and binary drift indicators. The ideal scalar correction coefficient is defined in equation (17) and is only used in the validation.

$$g_t^* = \text{clip} \left(\frac{\langle Y_t - \hat{Y}_t^B, \Delta_t \rangle}{\|\Delta_t\|_2^2}, 0, 1 \right). \quad (17)$$

This coefficient is learnt by a depth-2 histogram gradient-boosting regressor $h_\psi(z_t)$. The first 67% of validation origins fit the provisional gate; the remaining 33% select $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. The chosen configuration: 24 hour stride in the residual memory and $\lambda = 0.25$, the gate is then refitted, using the whole validation partition without test targets. The final forecast is given in equation (18).

$$\hat{Y}_t^{\text{TRACE}} = \hat{Y}_t^B + \lambda \text{clip}(h_\psi(z_t), 0, 1) \Delta_t. \quad (18)$$

Equations (17) and (18) neatly de-couple the validation supervision from test time inference; it is required that Y_t be present for training the gate, but it is not required to be present for all test reliability features.

E. Drift-Aware Forecast and Evidence Head

Dfeh is the common layer for output and audit in both direct feature model and TRACE-RAF (validation only drift router) and the frozen Chronos-Bolt benchmark. The proposed path is a combination of the context ensemble and gated residual update as illustrated in Figure 4. A separate branch to maintain M09 feature level conditioning and Chronos-Bolt forecast fusion to test their effect without them getting mixed up with TRACE. The drift scores and flags stay as reliability input, subset definitions, evidence fields; the last TRACE gate is set prior to the test partition being opened.

1) *Feature-Level Forecasting and Drift Diagnostics*: The direct model input and model prediction for horizon h is given by equation (19).

$$u_{t,h} = [q_t \parallel \rho(G_t) \parallel d_t \parallel c_{t+h}], \quad \hat{y}_{t+h}^X = f_h(u_{t,h}), \quad (19)$$

For the M09 comparison model, $u_{t,h} \in \mathbb{R}^{153}$ for each f_h , which is an XGBoost regressor trained with a squared-error objective. This design allows for having a different supervised function for every forecast horizon and a common origin representation. The equation (19) is valid for the proved 153 characteristics: out of them 85 context and 53 retrieval, 6 drift and 9 future -calendar value. The drift score and binary flag are only used for the evidence record and regime analysis and are not adaptive forecaster selection in the experiment reported here.

2) *Frozen-TSFM Forecast Fusion*: The retrieval variant, called the Chronos-Bolt, fuses the forecast level instead of using internal Channel Prompting. Suppose that the frozen forecast Chronos $\hat{Y}_t^C$ and the historical retrieved forecast $\hat{Y}_t^R$ exist. The fused prediction is defined as in equation (20).

$$\hat{Y}_t^{CR}(\omega) = \omega \hat{Y}_t^C + (1 - \omega) \hat{Y}_t^R, \quad (20)$$

where $\omega \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected $\omega = 0.25$. This branch just does not change the embedding or weights for Chronos-Bolt as those changes are made after both of these forecasts are made. Thus, (20) is not an internal TSFM prompting operation, it's an output space interpolation.

3) *Distribution-Shift Diagnostics*: The implementation does not claim verified concept-drift identification; it represents distribution shift using transparent indicators. These indicators are: recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift and event-burst ratio. The raw drift vector which is used by the code is defined by (21).

$$r_t = \left[|\mu_{24,t} - \mu_{168,t}|, \left| \log \frac{\sigma_{24,t} + \epsilon}{\sigma_{168,t} + \epsilon} \right|, 1 - \text{clip}(\bar{s}_{ts,t}, 0, 1), \sqrt{\frac{1}{4} \sum_{\ell=1}^4 \left(\frac{w_{t,\ell} - \tilde{w}_{\ell}^{tr}}{s_{w,\ell}^{tr}} \right)^2}, \max(b_{event,t}, 0) \right], \quad (21)$$

The weather term is calculated using standardized temperature, relative air humidity, pressure and wind speed (standardized using training medians and standard deviations). The robust scaling of each component $r_{t,j}$ is carried out in equation (22)

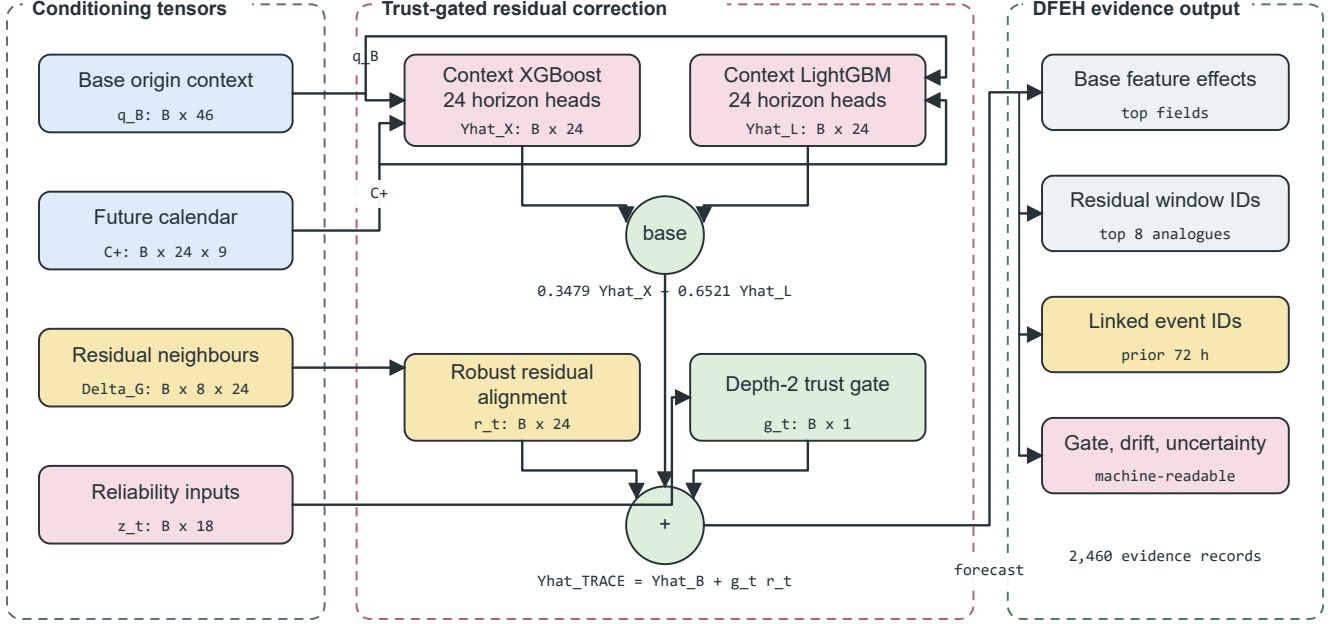


Fig. 4. TRACE-RAF and the Drift-Aware Forecast and Evidence Head. Out-of-fold LightGBM and XGBoost predictions form a context ensemble, LSER returns aligned residual analogues, and an 18-variable reliability gate scales the correction. The final forecast is accompanied by feature effects, retrieved cases, event identifiers, drift components, and uncertainty evidence.

with the training median and median absolute deviation (MAD).

$$d_{t,j} = \frac{1}{6} \text{clip} \left(\left| \frac{r_{t,j} - \text{med}_{tr}(r_j)}{\max(1.4826 \text{MAD}_{tr}(r_j), \epsilon)} \right|, 0, 6 \right). \quad (22)$$

The continuous score and binary flag is as defined in equation (23).

$$S_t = \frac{1}{5} \sum_{j=1}^5 d_{t,j}, \quad D_t = \mathbb{I}[S_t \geq Q_{0.90}(\{S_v : v \in \mathcal{T}_{val}\})]. \quad (23)$$

Equations (21) – (23) correspond to the detector implemented, with reference statistics set by data from the training set, the 0.90-quantile threshold determined by data from the validation set and the data from the test set transformed without refitting. The binary D_t is kept for diagnostic regime analysis, and for forecast evidence.

4) *Evidence Record and Uncertainty Proxy*: The evidence module leverages signed local effects from the 24 direct models, the weather flags, the top three retrieved windows, the recent event records, the calendar information, the forecast direction, and the drift evidence at the component level. It has its uncertainty proxy defined in equation (24).

$$u_t^{\text{proxy}} = \sqrt{\bar{\sigma}_{R,t}^2 + \bar{e}_{val}^2}, \quad (24)$$

with the mean horizon-wise spread of the retrieved trajectories ($\bar{\sigma}_{R,t}$) and the validation residual RMSE ($\bar{e}_{val}$) respectively. The outcome is a diagnostic scale, a calibrated prediction interval is not produced. Each evidence record is based on stored fields as opposed to an unrestricted language model

and each event statement is still not causal. Only evidence-field definition of equation (24) is retained and not for the construction of the confidence interval.

F. Relationship to TimeRAF

TimeRAF is a top- k retrieval model for time-series windows with an MLP dual encoder and Channel Prompting [9] to incorporate the candidate embeddings in the frozen TSFM. Table II lists the principles that remain the same and the different mechanisms. The comparison is not a performance ranking as the studies are based on different datasets, horizons and protocols.

VII. EXPERIMENTAL SETUP

This is the final experiment for the manifest with id 20260821T143600090063Z. All model, retrieval-density, fusion and threshold/gate decisions were made, before evaluating the final test partition, based on the training and validation partitions. You can find the source, configuration, predictions, tables, evidence records and figures used here in the archived package.

A. Dataset Description and Audit

The study period is 2019–2025. The PM2.5 measurements were obtained from the US EPA AirData/AQS parameter 88101 for Los Angeles County [38]. No single monitoring site was used as the target; the audited target is the hourly median across the official Los Angeles County AQS monitors. The

TABLE II
METHODOLOGICAL COMPARISON OF TIME-RAF AND TRACE-RAF

Aspect	TimeRAF	TRACE-RAF
Primary objective	Improve zero-shot forecasting by augmenting a frozen TSFM with external time-series knowledge.	Evaluate source-audited, event-context residual retrieval for local PM2.5 forecasting under temporal shift.
Forecasting domain	Multi-domain benchmark datasets.	Los Angeles County hourly PM2.5, 2019–2025.
Knowledge base	Curated, consistently normalized time-series windows.	1,824 primary training windows, dense-memory sweeps up to 45,784 windows, and 909 selected OOF residual candidates.
Retriever	Learned MLP dual encoder with forecaster-guided training.	Random, cosine, and transparent hybrid similarity; no learned retriever.
Knowledge integration	Internal Channel Prompting of candidate embeddings.	Trust-gated residual correction; feature- and forecast-level fusion are comparison arms.
Backbone	Frozen TSFM, principally TTM-Base.	Convex LightGBM–XGBoost context ensemble; frozen amazon/chronos-bolt-small is a benchmark.
Leakage control	Non-overlapping knowledge windows and cross-dataset training constraints.	Candidate target must end before query lookback; validation and test retrieval use training history only.
Event handling	Not an explicit module.	Availability-tagged NOAA Storm Events features and source-record evidence, reported retrospectively.
Drift and evidence	Retrieved-case inspection.	Five-component drift audit, reliability-gated residuals, signed feature effects, retrieved IDs, event IDs, and uncertainty evidence.
Implementation scope	Learned retriever and Channel Prompting are implemented.	Those mechanisms are not implemented and remain future extensions.

fallback chosen was 06-037-COUNTY and observed target hours were 61353, contributing source records were 191284, and the observed coverage was 0.9998. The PM2.5 value was always reported in micrograms/m3.

Weather covariates were taken from NOAA GHCNh annual archives [39]. The station chosen was 72297023129: Long Beach / Daugherty Field / Airport, 10.36 km away from the centroid of the target PM2.5. After a past-only short-gap filling, the complete weather-feature coverage was 0.9923 and raw measured coverage of the core was 0.9882. Every year had a coverage rate above 0.980 and the record lasted until 31 Dec 2025.

Event data were downloaded from the NOAA Storm Events annual detail archives [40] on an environmental event by event basis from those archives which were verified as a hash. As the direct access to the NCEI was not available at the execution time, the experiment used a local staged NCEI cache with unaltered NOAA annual archives as well as a source manifest consisting of official URLs, byte counts, and SHA-256 hashes. No changes were made to delivery mode `verified_official_noaa_cache` for the audit. The Event Audit identified 127 event days, 2,557 source-overlap days and four qualifying event categories. Records are broken down into the following categories: High Wind, Flood, Heavy Rain, Other Weather and Wildfire. NOAA Storm Events will not provide machine-readable publication times, so event-aware outputs will assume that the event occurred at the same time as the event was recorded. They are not hard deadline event forecasts! The output, readiness status, is presented in Table III. The source audit and window contract are shown in Figure 2.

Table IV summarizes the assessed scope and various sources’ contribution to the forecasting process.

The hourly source panel is reconciled to the model population in the origin-level attrition audit in Table V. The greatest loss is when all causal origin features are made to be finite, where 56,194 of 61,177 bounded candidate origins are kept

TABLE III
DATA-READINESS AUDIT

Gate	Value	Status
PM2.5 observed coverage	0.9998	Pass
Weather complete coverage	0.9923	Pass
Event days	127	Pass
Event overlap days	2,557	Pass
Qualifying event categories	4	Pass
Strict event availability	False	Caveat

(91.85%). Retained origin has 24 target values, all of which can be directly observed.

B. Data and Code Availability

There are raw publicly available public records and downloadable pages from EPA AirData, NOAA GHCNh, and NOAA Storm Events [38]–[40]. Since direct access to NCEI was not available in the experiment environment, a locally staged NOAA Storm Events cache was used, and all cached NOAA files were checked against the created manifest file of the sources. The source code is on github at <https://github.com/shasabbir/event-time-raf>. The implementation is done in Python 3.12.13, with the use of NumPy, pandas, scikit-learn, XGBoost, PyTorch and the `chronos-forecasting` package, and the repository includes the configuration, notebooks, source modules and tests that can be used to re-run the pipeline when public source files are available.

C. Forecasting Task and Split

The forecasting task makes a forecast prediction of PM2.5 between 1 hour to 24 hours in the future with 168 hours lookback. Training contains 45,784 origins from January 8, 2019 to August 23, 2024; validation contains 7,950 origins from August 25, 2024 to August 23, 2025 and the frozen test holdout contains 2,460 origins from August 25 to December

TABLE IV
DATASET AND SOURCE SUMMARY

Data group	Source	Evaluated scope	Use in pipeline
PM2.5	EPA AirData/AQS hourly parameter 88101	Los Angeles County, 2019–2025; 61,353 observed target hours	Target sequence and lag/rolling features
Weather	NOAA GHCNv, station 72297023129	2019–2025; 0.9923 usable core coverage	Temperature, humidity, wind, pressure, and rain features
Events	NOAA Storm Events annual detail archives	127 event days; four qualifying categories	Retrospective event-start features and forecast evidence
Calendar	Generated from timestamps and public holiday rules	2019–2025 hourly calendar	Hour, weekday, weekend, month, season, and holiday features

TABLE V
FORECAST-ORIGIN ATTRITION AUDIT

Stage	Origins	Retained (%)
Bounded candidate origins	61,177	100.00
Chronological split eligible	61,129	99.92
Complete 168-hour lookback	61,121	99.91
Complete observed target	60,953	99.63
Complete causal origin features	56,194	91.85
Final retained windows	56,194	91.85

30, 2025. The end of its targets is December 31, 2025, and the total number of evaluated hourly points is 59,040. Test targets are not used for model/threshold selection, and random splitting is not used.

D. Models

The baseline set includes: seasonal rules, regularized linear regression, boosted trees, neural sequence models, retrieval-only forecasts and a frozen TSFM. The proposed M13 TRACE-RAF model is contrasted to an ensemble of its C06 context model and the event-free A03 gate. M12 is a drift router, it has been separately validation selected, it is not a post-hoc product. Chronos placebos combine the same frozen TSFM forecast with climatology and persistence and take the forecast averaging away from retrieval. The amazon/chronos-bolt-small foundation model is used for the evaluation. The evaluated configurations are listed by their forecasting and retrieval tasks in Table VI.

E. Implementation and Hyperparameters

The experiment was performed on a cloud-based Linux 6.12.90, Python 3.12.13. The publication pipeline has measured 6,915.3 seconds of its internal run time. The following are the key package versions that are recorded in the run manifest: NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, LightGBM 4.6.0, PyTorch 2.10.0+cu128, Transformers 4.57.6, and chronos-forecasting 2.3.1. The main hyperparameters are given in table VII.

F. Evaluation Metrics and Statistical Checks

The main measurements are: MSE, MAE, RMSE, MAPE, sMAPE and R^2 . MSE is kept as the main squared error metric and MAE and RMSE give more easily interpreted PM2.5

error scales. The following are the main error measures for n observations y_j and predictions $\hat{y}_j$.

$$\begin{aligned} \text{MSE} &= \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2, \\ \text{MAE} &= \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|, \\ \text{RMSE} &= \sqrt{\text{MSE}}, \end{aligned} \quad (25)$$

The explained variance is summarized in equation (26).

$$R^2 = 1 - \frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{\sum_{j=1}^n (y_j - \bar{y})^2}. \quad (26)$$

Both equation (25) and (26) are aggregated for all forecast origins and horizons to arrive at the overall result. The PM2.5 percentage errors (MAPE and sMAPE) are stored in the machine-readable output but are not used in the main ranking as percentage errors could be prone to instability at low PM2.5 values. Subset metrics are only displayed if there are 50 or more forecast origins. For the 72 hour event-context subset, 70 origins are included; for the drift subset, 407 origins are included, and for the 33 target-event and eight active-event origins, they are included in counts without being displayed in performance tables. Pairwise comparisons are based on 5,000 paired moving-block bootstrap resamples with 168-hour blocks, and Diebold–Mariano tests with 168 HAC lags. The p -values are stored both before and after Holm transformation. The thresholds for decisions based on the AQI are considered at 35.4 and 55.4 $\mu\text{g}/\text{m}^3$, while the thresholds for decisions based on the Chronos quantiles are considered by interval coverage, approximate quantile-grid CRPS, and calibration error.

G. Comparison with Existing Studies

The results cannot be compared directly with those of previous TSFM and air-quality papers as they rely on different data sets, forecast periods, variables and evaluation methods. Thus, Table VIII is a methodological coverage comparison, not an attempt to achieve a cross-paper performance dominance.

VIII. RESULTS AND DISCUSSION

A. Overall Comparison with Baselines

The final-holdout results are given in Table IX. The Ridge regression with context features yields the best performance

TABLE VI
EVALUATED MODEL FAMILIES

ID	Description
C00–C06	Hour–month climatology; Ridge and LightGBM context; calendar and event retrieval; raw-event XGBoost; convex context ensemble
M00–M02	Persistence, daily seasonal, and weekly seasonal naive baselines
M03–M09	PM2.5/context XGBoost; retrieval-only controls; feature-level retrieval, event, and drift variants
B00–B02	DLinear, PatchTST, and LSTM trained on the same PM2.5 windows
M10–M12	Frozen Chronos-Bolt, Chronos retrieval fusion, and validation-only drift router
M13	Proposed TRACE-RAF trust-gated residual analogue correction
A00–A03	Event-free, random-retrieval, matched feature-count placebo, and event-free TRACE ablations
P00–P01	Chronos fused with climatology or persistence placebos

TABLE VII
EXPERIMENTAL HYPERPARAMETERS AND RUNTIME SETTINGS

Component	Setting
Study period	2019–2025
Lookback and horizon	$L = 168$ hours, $H = 24$ hours
Split	45,784 train; 7,950 validation; 2,460 final-test origins
SACB feature groups	23 PM2.5, 14 weather, 9 calendar, 39 event features
SACB missing-data rule	Past-only filling for gaps up to 3 hours; no target interpolation
LSER retrieval	$k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$
LSER knowledge base	1,824 windows at 24-hour primary stride; 6/1-hour sweeps
LSER similarity weights	time series 0.5, weather 0.2, calendar 0.1, event 0.2
DFEH direct input	153 features for each of 24 horizon regressors
DFEH XGBoost	350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$
TRACE base ensemble	LightGBM weight 0.6521; XGBoost weight 0.3479
TRACE residual memory	OOB boundaries 0.50/0.75/1.00; 909 selected candidates
TRACE trust gate	18 inputs; depth 2; 100 iterations; learning rate 0.05; selected strength 0.25
DFEH drift detector	5 components; 24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold
DFEH Chronos path	amazon/chronos-bolt-small, batch size 64
DFEH fusion weights	$\{0, 0.25, 0.5, 0.75, 1.0\}$; selected $\omega = 0.25$
Neural baselines	batch 256; at most 30 epochs; patience 5; learning rate 0.001
Forecast evidence record	Top 3 feature effects, windows, and recent event IDs
Inference	5,000 paired resamples; 168-hour blocks/HAC lags; Holm correction

TABLE VIII
METHODOLOGICAL COMPARISON WITH RELATED FORECASTING WORK

Method family	TSFM	Retrieval	Events	Evidence	Comment
PatchTST-style Transformers [24]	No	No	No	Limited	Strong sequence modeling, but event context is not central.
Chronos [5]	Yes	No	No	Limited	Frozen zero-shot baseline used in this study.
TimeRAF [9]	Yes	Yes	No	Retrieval cases	Base retrieval-augmented TSFM method; does not target official event-context PM2.5.
XGBoost context baseline [10]	No	No	No	Feature importance	Strong supervised tabular baseline for local PM2.5.
TRACE-RAF	No ^a	Yes	Yes	Evidence-grounded	Adds official-source audit, OOB residual retrieval, a trust gate, drift diagnostics, and stored evidence.

^aChronos-Bolt is evaluated as a frozen benchmark, but the proposed TRACE-RAF forecast is a supervised residual-correction model rather than a foundation model.

overall with MSE 39.144, MAE 4.079, RMSE 6.257 and $R^2 = 0.457$. TRACE-RAF obtains MSE 40.062, MAE 4.134, and RMSE 6.329. It outperforms M04 XGBoost context by 4.63% in terms of MSE, and is within 0.07% of C05 LightGBM context, but does not outperform Ridge or LSTM. This is an important ranking as it indicates that the proposed correction is competitive with nonlinear context models, and the smooth county-median target and late-2025 holdout are still favourable to regularized linear structure.

The frozen Chronos-Bolt evaluation went without problem and the validation procedure chose a TSFM weight of 0.25.

Event-retrieval fusion increases MSE from 47.534 to 48.609 and MAE from 4.202 to 4.574. The increase in MAE of 0.372 is still significant after Holm adjustment (bootstrap $p = 0.038$; DM $p = 0.029$). Furthermore, the persistence-fusion placebo is not as high as M10 or M11 (MSE 46.964). So this run also provides evidence against the possibility of a generic averaging effect of event retrieval. Chronos’ native quantiles have useful diagnostic numbers: for M10, 3.187 for CRPS, 0.017 for mean absolute calibration error, and an empirical coverage rate of 0.761 for the nominal coverage rate of 0.80, while for M11 the numbers are 4.120, 0.176, and 0.206.

TABLE IX
MANIFEST-BACKED OVERALL TEST RESULTS FOR 24-HOUR PM2.5 FORECASTING

Model	MSE	MAE	RMSE	R^2
C00 hour-month climatology	73.915	6.398	8.597	-0.025
M00 persistence	57.327	4.840	7.571	0.205
M03 XGBoost PM2.5	44.547	4.317	6.674	0.382
M04 XGBoost context	42.006	4.233	6.481	0.417
C01 Ridge context	39.144	4.079	6.257	0.457
C05 LightGBM context	40.089	4.151	6.332	0.444
B00 DLinear	40.305	4.097	6.349	0.441
B01 PatchTST	40.766	4.111	6.385	0.435
B02 LSTM	39.978	4.095	6.323	0.445
C06 context ensemble	40.115	4.156	6.334	0.444
M09 event-feature XGBoost	41.193	4.148	6.418	0.429
M13 TRACE-RAF	40.062	4.134	6.329	0.444
A03 TRACE-RAF without events	40.062	4.135	6.329	0.444
M10 frozen Chronos-Bolt	47.534	4.202	6.894	0.341
M11 Chronos + event retrieval	48.609	4.574	6.972	0.326
M12 validation drift router	41.466	4.190	6.439	0.425

B. Ablation Study

The pre-declared paired MSE contrasts are reported in Table X, where negative differences are in favor of the first model. The cosine retrieval features are nominally better than M04, while raw event features nominally outperform the matched feature-count placebo. The MSE of TRACE-RAF is almost the same as both of its ensembler and LightGBM. Its MAE differences are -0.021 versus C06 (unadjusted 95% block-bootstrap CI $[-0.038, -0.005]$) and -0.017 versus C05 (unadjusted 95% block-bootstrap CI $[-0.031, -0.001]$), but their Holm-adjusted bootstrap p -values are 0.281 and 0.394. Comparing the M13–A03 contrast to the TRACE-RAF overall effect reveals that it is extremely close to zero, which directly demonstrates that TRACE-RAF’s overall effect is not explained by event conditioning. In this archived run, Holm correction was applied over the complete 25-comparison family for each metric. The only MAE contrast that is adverse and significant after Holm correction is M11–M10.

The accuracy does not increase monotonically with the ablation. The contributions of the validation faithfulness are dominated by the PM2.5 history, weather and calendar features while the event, retrieval and drift groups contribute less. The paper, as a result, ascribes TRACE-RAF’s actions to selective residual correction, instead of a well-documented general effect.

The k sensitivity for retrieval-only cosine baseline in Table XI shows that the more the number of neighbors, the better the retrieval-only baseline within the range tested. The best retrieval-only configuration among $k \in \{1, 4, 8, 16\}$ is $k = 16$, with MSE 51.316. The reported feature models were with the default $k = 8$.

Knowledge-base density shows a non-monotone behaviour. As can be seen from Table XII, both the retrieval-only and the feature-level event-aware versions provide the best 6-hour memory: it is not optimal to select all hourly candidates. Event-conditioned retrieval is better than cosine retrieval by 1.669 MSE at 6 hour stride while it is worse at 24- and 1- hours strides. This interaction is an impetus for reliability gating but not a statement that event similarity automatically is desirable.

C. Event and Drift Subsets

Table XIII gives the two pre-specified subsets which satisfy the 50-origin gate. A target-event overlap (direct) includes only 33 origins, in the broader 72-hour event-context subset 70 origins. The mean of the event-context targets is 6.26, and the variance is 11.26; and the mean of the targets outside the subset is 11.14, and variance 73.20. Thus, the magnitudes of cross-subset MSE are not interpreted as “difficulty of the event”. TRACE-RAF and its event-free ablation are indistinguishable on the same event-context origins. The drift subset has 407 origins that were selected by the DFEH threshold, which was validated/calibrated, and has much higher target variance (105.34), which contributes to the higher absolute error.

The composite detector is able to detect 407 test origins with threshold value of 0.2132, and the validation-calibrated two-sample KS benchmark is able to detect 348 test origins with threshold value of 0.4523. Their Jaccard agreement is only 11.7%. The composite flag is thus considered to be a model-specific regime label and not necessarily a definition of drift. It is reinforced by a separately selected M12 router, which is nominally better with M04 but does not pass Holm correction.

D. Site-Level and Decision Sensitivity

The median for the county is the key target value, since none of the individual monitors achieve the 0.70 coverage target. To check if the event finding is only based on aggregation, three arms at the monitor level were trained but the temporal protocol not changed. As shown in Table XIV, effects are heterogeneous, event retrieval is impaired at site 1103, and is somewhat improved at sites 4009 and 9035. Event-only site subsets are made up of 10-33 origins and are not scored. These are internal and not external geographic target-sensitivity checks.

There are only 37 exceedances among 2,460 test origins for the $35.4 \mu\text{g}/\text{m}^3$ 24-hour-mean threshold with no origin exceeding the $55.4 \mu\text{g}/\text{m}^3$ threshold. The recall of physical-threshold forecasts is very low. Right now, LightGBM achieves 0.234 for

TABLE X
SELECTED 168-HOUR PAIRED BLOCK-BOOTSTRAP ABLATIONS

Comparison	MSE diff.	Unadj. 95% CI low	Unadj. 95% CI high	Holm-adjusted p
M04 – M03: weather/calendar	-2.542	-6.695	0.940	1.000
M07 – M04: cosine retrieval features	-1.071	-2.280	-0.041	1.000
M08 – M07: event-conditioned ranking	0.151	-0.447	0.789	1.000
C04 – A02: event signal/placebo	-0.524	-0.983	-0.142	0.389
M09 – M08: drift features	0.107	-0.386	0.502	1.000
M13 – C06: TRACE residual gate	-0.052	-0.393	0.353	1.000
M13 – A03: TRACE event channel	-0.0001	-0.006	0.008	1.000
M13 – C05: proposed/best tree	-0.027	-0.350	0.353	1.000
M11 – M10: Chronos retrieval	1.075	-7.177	6.712	1.000
M12 – M04: drift router	-0.540	-1.112	-0.066	0.747

TABLE XI
COSINE RETRIEVAL SENSITIVITY TO k

k	MSE	MAE	RMSE
1	84.602	5.868	9.198
4	57.092	5.011	7.556
8	53.549	4.860	7.318
16	51.316	4.773	7.164

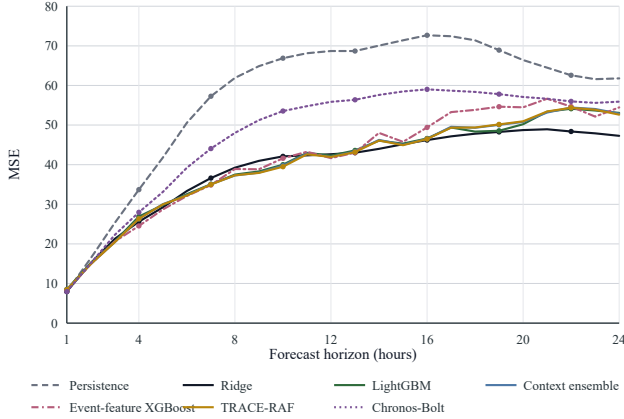


Fig. 5. MSE by forecast horizon on the Los Angeles County test set.

precision, 0.405 for recall, and 0.297 for F1, while TRACE-RAF achieves 0.213, 0.351, and 0.265 respectively with the same. Sparse-event results are reported to link these results to air-quality decisions, but these results are not adequate for a stand-alone alerting claim.

E. Horizon-Wise and Retrieval Diagnostics

Horizon-wise test errors for comparators Principal supervised, feature-retrieval, frozen-TSFM and TRACE-RAF are displayed in Fig. 5 and Fig. 6, respectively. None of the models have a highly localized failure in the late horizon which could render the overall comparison false. The DFEH score and fixed threshold are plotted on the test time line in Fig. 7. Figures 8 and 9 are just qualitative checks, they are not used to select models. All plots are regenerated from run 20260821T143600090063Z.

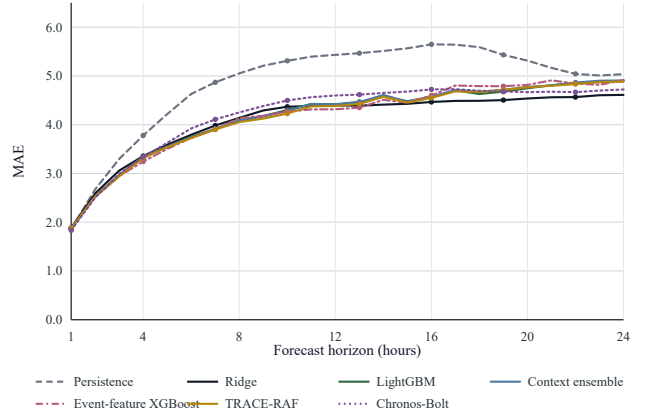


Fig. 6. MAE by forecast horizon on the Los Angeles County test set.

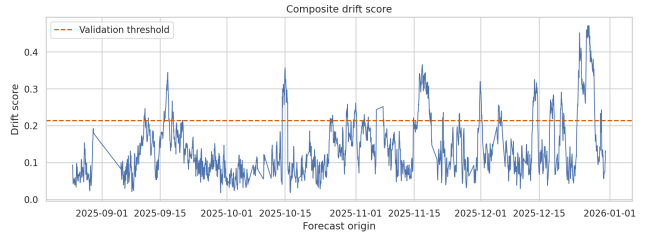


Fig. 7. Composite drift score and the validation-calibrated threshold on the final holdout. The flag defines a diagnostic regime rather than verified distribution-shift ground truth.

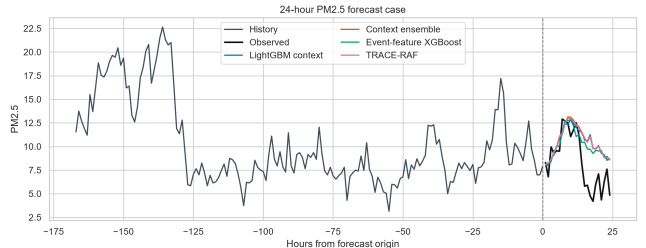


Fig. 8. Representative 24-hour forecast from the final holdout. Model traces are plotted against the observed hourly target without post-test adjustment.

F. Forecast Evidence and Auditability Analysis

In the context of DFEH, a source-grounded forecast evidence record is retained rather than claiming a complete causal

TABLE XII
KNOWLEDGE-BASE DENSITY SENSITIVITY ON THE FINAL HOLDOUT

Stride (h)	Candidates	Cosine MSE	Event-retrieval MSE	M08 MSE	M09 MSE
24	1,824	53.549	54.098	41.085	41.193
6	7,608	49.524	47.855	40.520	40.886
1	45,784	52.738	54.253	41.172	41.749

TABLE XIII
REPRESENTATIVE EVENT AND DRIFT SUBSET RESULTS

Subset	Model	Origins	MSE	MAE
Event context	C01 Ridge context	70	5.624	1.925
Event context	C06 context ensemble	70	6.568	2.101
Event context	M13 TRACE-RAF	70	6.552	2.097
Event context	A03 TRACE without events	70	6.553	2.095
Non-drift	C01 Ridge context	2,053	38.808	3.989
Non-drift	M13 TRACE-RAF	2,053	38.085	3.955
Drift	C01 Ridge context	407	40.843	4.536
Drift	C06 context ensemble	407	50.267	5.077
Drift	M13 TRACE-RAF	407	50.033	5.039
Drift	A03 TRACE without events	407	50.047	5.039

TABLE XIV
MONITOR-LEVEL EVENT-RETRIEVAL SENSITIVITY

Site	Coverage	Origins	M04 MSE	M08 MSE
06-037-1103	0.558	1,631	68.281	69.898
06-037-4009	0.510	1,732	73.506	73.095
06-037-9035	0.374	547	17.682	17.371

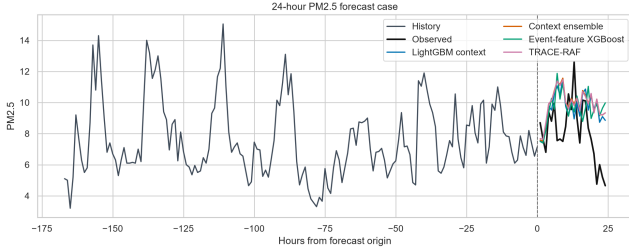


Fig. 9. Representative forecast whose target lies within the declared 72-hour event-context window. The event designation is retrospective because Storm Events files do not expose publication timestamps.

explanation. The run and window identifiers, origin time, retrieved historical windows, linked event records, leading feature effects, drift components, retrieval spread, validation residual RMSE, uncertainty proxy and drift flag, and the event-availability mode are stored in the pipeline for each of the 2,460 test origins. The resulting `explanations.parquet` file thus is in 1:1 relation with the forecasts evaluated. Its main local fields are typically current PM2.5 and recent rolling concentrations and analogue and event identifiers provide exposure of what evidence is available to the correction mechanism.

The information of the evidence layer is summarized in Table XV. Such a design is conducive to auditability: all evidence statements are mapped to a model feature, a windowed retrieval time-series or an official event record, or one of the drift components. This does not prove causality: evidence of

TABLE XV
FORECAST EVIDENCE STORED BY TRACE-RAF

Evidence field	Interpretation role
Top feature effects	Local model drivers from the direct fore-caster.
Retrieved windows	Historical analogues used by the retrieval module.
Event evidence IDs	Official storm-event context linked to the origin.
Drift components	Mean, variance, weather, similarity, and event-burst shift signals.
Uncertainty proxy	Combined retrieval spread and validation residual error.

TABLE XVI
M09 FEATURE-FAMILY FAITHFULNESS PERTURBATIONS

Feature family	Permutation Δ MSE	Median deletion Δ MSE
PM2.5 history	25.575	13.531
Weather	3.440	3.089
Calendar	3.548	2.481
Events	1.162	-0.442
Retrieval	0.964	-0.551
Drift	0.268	0.125

events is reported as contextual support and the drift flag is reported as an uncertainty warning. Faithfulness is assessed by varying complete sets of model features in the fitted M09 model. Table XVI shows that PM2.5 history is by far the most influential and weather and calendar information are still influential. Event and retrieval permutations increase MSE while negative effects on MSE are found for the median-deletions. Because of these mixed signs it is not possible to make a stronger claim that every stored evidence type is a stable causal explanation.

G. Computational Cost and Reproducibility

The publication pipeline had internal runtime of 6,915.3 seconds in a Linux environment running Python 3.12.13. All

TABLE XVII
COMPUTATIONAL AND REPRODUCIBILITY SUMMARY

Item	Recorded value
Run identifier	20260821T143600090063Z
Pipeline runtime	6,915.3 seconds
Platform	Linux 6.12.90
Python	3.12.13
Forecast horizon	24 hours
Lookback window	168 hours
Bootstrap resamples	5,000 in 168-hour blocks
Multiple testing	Holm adjustment over the comparison family
Chronos checkpoint	amazon/chronos-bolt-small
Code availability	https://github.com/shasabbir/event-time-raf

three notebooks completed with no cell errors. The recorded versions include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, LightGBM 4.6.0, PyTorch 2.10.0+cu128, Transformers 4.57.6, and `chronos-forecasting` 2.3.1. There are 42 result tables in the archive with a single run identifier, and they include configuration and artifact hashes, prediction-level outputs and evidence records, as well as an independent claim-verification notebook. The reconstruction of the TRACE subset-inference is in agreement with the stored prediction with a maximum absolute difference of 9.85×10^{-17} . No hardware identifiers or Git commit were found in the notebook environment being extracted, hence runtime is not considered as hardware-normalized efficiency result.

The primary interpretation of this is that supervised context models are particularly good in this Los Angeles County task. TRACE-RAF performs nominally better than the LightGBM comparator in both MSE and MAE, with Ridge overall being the best, but no favorable TRACE contrast survives family-wise adjustment. The similar event-free ablation also demonstrates that the collective TRACE outcome isn’t the result of measurable event conditioning, but rather conservative residual correction. However, event-conditioned output-space fusion with frozen Chronos-Bolt-small has a distinctly negative effect on the MAE after Holm correction. The outcome is thus a reproducible boundary: trust-gated residual analogue correction is competitive, while this frozen Chronos-Bolt-small output-fusion configuration is not supported in the holdout. This result does not evaluate internal TSFM prompting or retrieval-augmented TSFMs generally.

There are 622 origins in a pre-specified January 2025 stress analysis, but it is within the validation period, and is only reported as development evidence. Ridge, Chronos-Bolt, the context ensemble, and TRACE-RAF obtain MSE 118.300, 116.012, 123.305, and 123.643, respectively. It’s not an extra test set, and will not affect the final model rank.

Lastly, every event aware interpretation should be interpreted with the caveat of availability. The source has official and hash-verified NOAA Storm Events, but does not publish machine-readable publication timestamps. Event-related results are retrospective sensitivity results, with the experiment reporting the availability time as the time that the events began. A true operation study would involve event or smoke sources with “true” issue or publication dates, such as an audited

NOAA HMS cache or alert product archive.

IX. LIMITATIONS

First is the availability of events. NOAA Storm Events are official and are hash verified, and the public detail files do not give machine-readable publication times. For the SACB, the start-date of an event is thus a proxy for availability and event-conditioned results are retrospective sensitivity analyses and not operational forecasts.

The second restriction is the geographic validation. The experiment is for 1 county-level PM2.5 aggregate for 2019-2025. Based on the monitor-level analysis, the event-retrieval effects change sign between sites, although transfer to another county, climate and/or emissions regime cannot be determined.

The third restriction is on what is in the holdout. A total of 33 target-event origins and 37 exceedances of thresholds are detected, and no test origin exceeds $55.4 \mu\text{g}/\text{m}^3$. The high error period in January 2025 is a period of validation and not external confirmations. Reporting of polluted smoke episodes or of operating alerts would thus be greater than the test support observed.

The fourth limitation is the construction of the targets. A County median makes it less likely that monitors are missing, but also less likely to show local peaks and is subject to the monitors who are active. The low coverage monitor arms detect some heterogeneity but are not dense enough to make inferences about events that comprise a subset of the measured events.

The fifth restriction is related to the empirical strength. The performance of TRACE-RAF is not superior to that of Ridge and both the residual-gate contrast and the event-channel contrast are not significant after Holm adjustment. Records of evidence are auditable, but the faithfulness perturbations of the event retrieval families are mixed and have no causal interpretation.

The sixth limitation is with regards to model integration. LSER is not a learned dual encoder, Chronos-Bolt is not an internal prompting backbone, and LSER is a transparent similarity retriever. The method should not be given as a copy of the TimeRAF Channel Prompting mechanism.

Lastly, hardware identifiers, peak memory and a git commit were not present within the run environment extract. Package versions, configuration hashes, archive hashes, and prediction-level verification provide better reproducibilities, and the execution time reported is not a hardware-normalized efficiency measure.

X. FUTURE WORK

Operational event studies should use archive that provides actual time of issue/publication instead of using event-start proxies and should include smoke-specific sources if the event transport of interest is for a wildfire. External validation to be carried out in a different US county with different meteorology and emissions and monitor density. Event effects then could be tested using spatial or site-aware forecasting. Future research could assess a learned residual retriever, richer

embeddings of events, internally-conditioned TSFMs and calibrated exceedance probabilities. For each extension, a pre-registered external holdout must be retained and component-wise ablation should be carried out under the embargo of time.

XI. CONCLUSION

In this study a method named as TRACE-RAF is presented for the auditable PM2.5 forecasting for a 24-hour period with introducing a trust gating approach for the residual analog error. SACB provides the harmonisation of the official PM2.5, GHCNh weather, calendar and Storm Events records, while LSER maintains a strict historical embargo; TRACE retrieves out-of-fold residual analogues and applies a learned reliability gate; and DFEH keeps prediction-level evidence and drift diagnostics. On the final 2025 holdout of 2,460 origins, TRACE-RAF obtains MSE 40.062, MAE 4.134, RMSE 6.329, and $R^2 = 0.444$. This boosts context XGBoost and is nominally better than the LightGBM comparator and Ridge is still best at MSE 39.144. None of the favorable TRACE comparisons pass Holm adjustment and the event-free TRACE ablation is virtually identical. The supported contribution is therefore a reproducible trust-gated residual-retrieval architecture and an auditable evaluation of event-conditioned residual retrieval and one frozen Chronos-Bolt-small output-fusion configuration. It is not a general claim about retrieval-augmented TSFMs or universal baseline superiority. Event issue times, external geographic testing and smoke-specific event validation will still be required prior to operational deployment.

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